Geographical and geological GIS boundaries of the Tibetan Plateau and adjacent mountain regions

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Introduction

Geographical scale, in terms of spatial extent, provide a basis for other branches of science. This dataset contains newly proposed geographical and geological GIS boundaries for the **Pan-Tibetan Highlands** (new proposed name for the High Mountain Asia), based on geological and geomorphological features. This region comprises the **Tibetan Plateau** and three adjacent mountain regions: the **Himalaya**, **Hengduan Mountains** and **Mountains of Central Asia**, and boundaries are also given for each subregion individually. The dataset will benefit quantitative spatial analysis by providing a well-defined geographical scale for other branches of research, aiding cross-disciplinary comparisons and synthesis, as well as reproducibility of research results.

The dataset comprises three subsets, and we provide three data formats (.shp, .geojson and .kmz) for each of them. Shapefile format (.shp) was generated in ArcGIS Pro, and the other two were converted from shapefile, the conversion steps refer to 'Data processing' section below. The following is a description of the three subsets:

(1) The GIS boundaries we newly defined of the Pan-Tibetan Highlands and its four constituent sub-regions, i.e. the Tibetan Plateau, Himalaya, Hengduan Mountains and the Mountains of Central Asia. All files are placed in the "Pan-Tibetan Highlands (Liu et al._2022)" folder.

(2) We also provide GIS boundaries that were applied by other studies (cited in Fig. 3 of our work) in the folder "Tibetan Plateau and adjacent mountains (Others' definitions)". If these data is used, please cite the relevent paper accrodingly. In addition, it is worthy to note that the GIS boundaries of Hengduan Mountains (Li et al. 1987a) and Mountains of Central Asia (Foggin et al. 2021) were newly generated in our study using Georeferencing toolbox in ArcGIS Pro.

(3) Geological assemblages and characters of the Pan-Tibetan Highlands, including Cratons and micro-continental blocks (Fig. S1), plus sutures, faults and thrusts (Fig. 4),

are placed in the "Pan-Tibetan Highlands (geological files)" folder.

Note: **High Mountain Asia**: The name 'High Mountain Asia' is the only direct synonym of Pan-Tibetan Highlands, but this term is both grammatically awkward and somewhat misleading, and hence the term 'Pan-Tibetan Highlands' is here proposed to replace it. **Third Pole**: The first use of the term 'Third Pole' was in reference to the Himalaya by Kurz & Montandon (1933), but the usage was subsequently broadened to the Tibetan Plateau or the whole of the Pan-Tibetan Highlands. The mainstream scientific literature refer the 'Third Pole' to the region encompassing the Tibetan Plateau, Himalaya, Hengduan Mountains, Karakoram, Hindu Kush and Pamir. This definition was surpported by geological strcture (Main Pamir Thrust) in the western part, and generally overlaps with the 'Tibetan Plateau' *sensu lato* defined by some previous studies, but is more specific.

More discussion and reference about names please refer to the paper. The figures (Figs. 3, 4, S1) mentioned above were attached in the end of this document.

Data processing

We provide three data formats. Conversion of shapefile data to kmz format was done in ArcGIS Pro. We used the *Layer to KML* tool in Conversion Toolbox to convert the shapefile to kmz format. Conversion of shapefile data to geojson format was done in R. We read the data using the *shapefile* function of the raster package, and wrote it as a geojson file using the *geojson_write* function in the geojsonio package.

Version

Version 2022.1.

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Citation

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Figures mentioned in the 'Introduction' section:

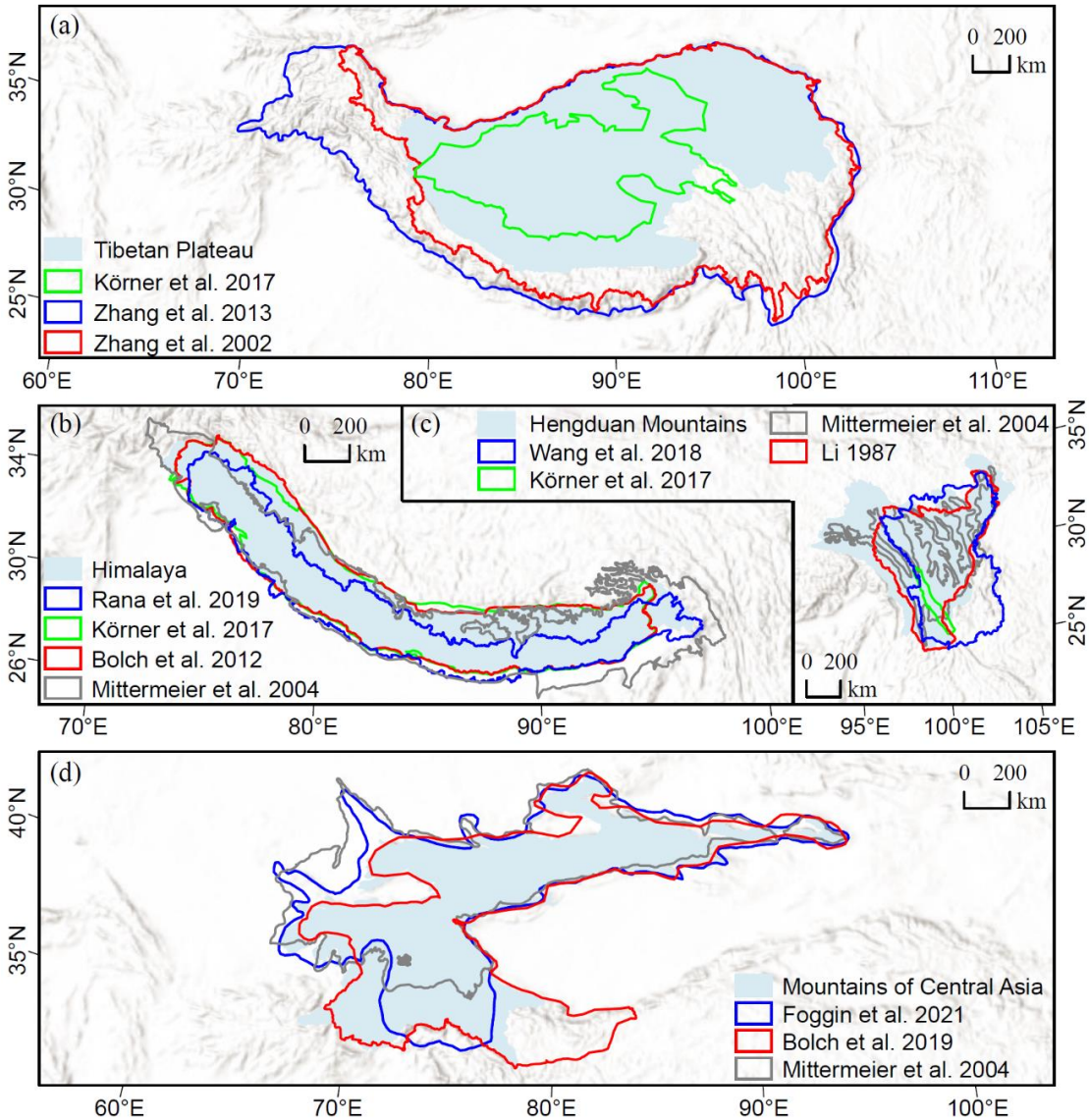

Fig. 3 Various geographical extent definitions for the Tibetan Plateau and adjacent mountains according to different sources, with the light blue-shaded region showing the newly defined area in this study. (a) Tibetan Plateau, attributed a smaller area according to , a medium sized area mainly confined to China, and a larger area defined as all land above 2,500 m. (b) Himalaya as defined according to Global Mountain Biodiversity Assessment (GMBA), a glacial study, plant diversity research and as a global biodiversity hotspot. (c) Hengduan Mountains as defined according to GMBA, ecosystem service research (Wang et al., 2018), geomorphology, and as a global biodiversity hotspot . (d) The extent of the Mountains of Central Asia as designated by (Mittermeier et al., 2004), (Bolch et al., 2019) and (Foggin et al., 2021).

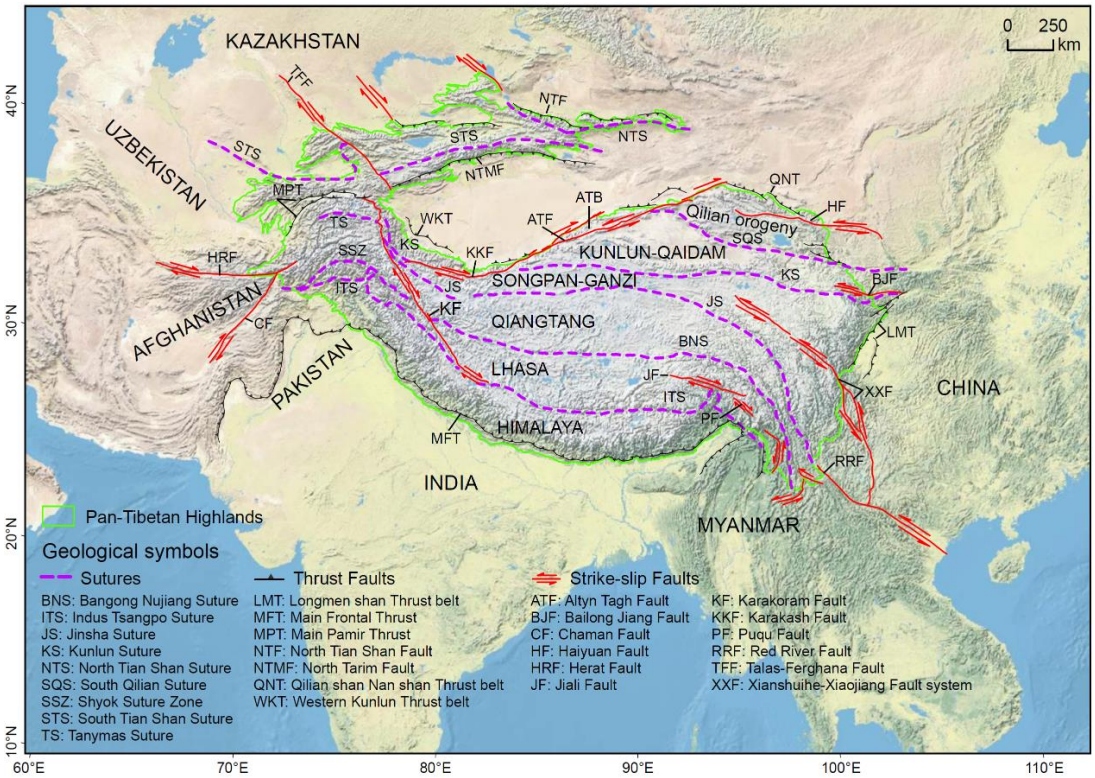

Fig. 4 Simplified regional tectonic map of the Pan-Tibetan Highlands (modified according to (McRivette et al., 2019; Ren et al., 2013; Taylor et al., 2009; Yin et al., 2000). Information on main sources of active faults is in accordance with (Jolivet et al., 2013; Taylor et al., 2009). Thrust faults have barbs on the upper plate; arrows indicate direction of movement of strike-slip faults. Major Mesozoic suture zones are illustrated by dashed purple lines. Capitalized words, Himalaya, Lhasa, Qiangtang, Songpan Ganzi and Kunlun-Qaidam, represent the main terranes. The Pan-Tibetan Highlands boundary is indicated by a green line. Base map is from the National Geographic Society.

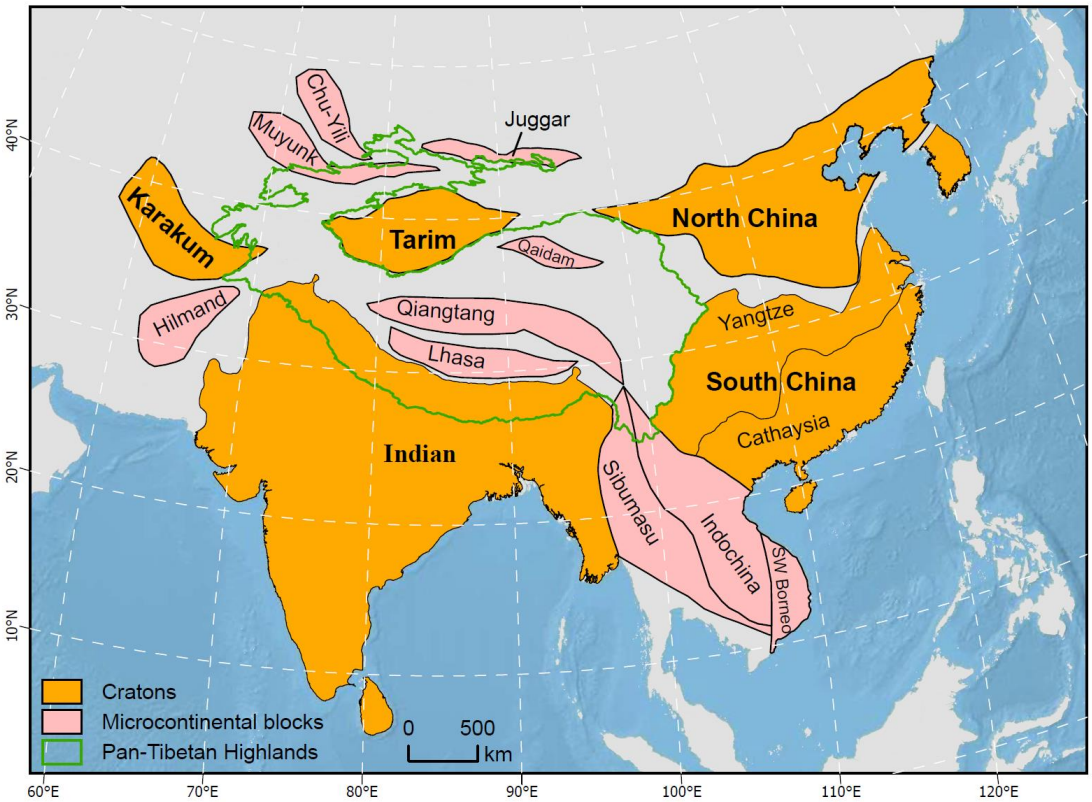

Fig. S1 Simplified tectonic map of the Tibetan Plateau and adjacent regions (redraw according to (Cawood et al., 2018; Li, 2006; Ren et al., 2013). The main cratons include North China (Sino-Korea), South China, India, Tarim and Karakum, while microcontinental blocks consist of Qaidam, Qiangtang together with adjacent blocks. The Pan-Tibetan Highlands boundary is shown by green line.